

PHANENDRA PALADUGU

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PROFESSIONAL SUMMARY

Data Scientist with 5+ years of hands-on experience designing, building, and deploying machine learning models, deep learning pipelines, and statistical models using Python, TensorFlow, Scikit-learn, and PySpark across enterprise and financial environments. Strong expertise in exploratory data analysis, feature engineering, predictive modeling, NLP, and computer vision. Experienced in building end-to-end ML workflows from data preprocessing and feature selection through model training, hyperparameter tuning, evaluation (AUC-ROC, precision/recall, F1), and production deployment with drift monitoring. Skilled in fine-tuning LLMs with QLoRA and GRPO, building RAG pipelines, and publishing models to HuggingFace for community use.

EXPERIENCE

Data Scientist | Wells Fargo (Indiana)

Jan 2025 - Present

- Build and deploy predictive models for credit risk assessment and customer churn prediction using XGBoost, LightGBM, and Scikit-learn on large-scale financial datasets, improving early default identification accuracy.
- Perform feature engineering and exploratory data analysis on transactional, behavioral, and demographic data using Pandas, NumPy, and PySpark to identify key risk predictors and customer segments.
- Develop model evaluation frameworks incorporating precision/recall, AUC-ROC, lift charts, and confusion matrices; track all experiments using MLflow for reproducibility and version control.
- Build automated data preprocessing and feature extraction pipelines using PySpark and SQL, handling missing values, outlier detection, encoding, and feature scaling on million-row datasets.
- Create interactive dashboards and model performance reports using Streamlit and Power BI to communicate risk scores, forecast trends, and analytical findings to risk management teams.
- Implement model monitoring and drift detection workflows using Evidently AI to track prediction stability, data distribution shifts, and trigger automated retraining alerts.
- Perform A/B testing and hypothesis testing on customer engagement strategies using statistical methods (t-tests, chi-square, bootstrap) to validate business decisions.
- Collaborate with risk management, compliance, and product teams to translate business requirements into data-driven strategies, presenting model outputs to senior stakeholders.

Lab Assistant | Purdue University (Indiana)

Aug 2024 - Dec 2024

- Assisted students in performing electrical measurements including voltage, current, impedance, and frequency response using oscilloscopes, multi-meters, and signal generators across multiple lab scenarios.
- Guided students in designing, wiring, and troubleshooting electronic circuits, applying systematic debugging approaches to identify and resolve hardware issues.
- Reviewed and evaluated lab notebooks and technical reports, providing detailed feedback on data accuracy, documentation quality, and analytical reasoning.

Technology Analyst | Infosys (India)

Jan 2020 - Dec 2021

- Designed and deployed a deep learning classification model using Python (TensorFlow/Keras) to auto-categorize 200+ monthly ServiceNow incidents and recommend resolution articles via NLP-based text similarity, reducing manual triage effort by 30%.
- Performed exploratory data analysis and feature engineering on large-scale enterprise datasets using Pandas, NumPy, and PySpark, applying statistical tests and correlation analysis to identify key predictors.
- Built predictive models using Scikit-learn and XGBoost for demand anomaly detection across supply chain data, enabling early identification of disruptions and improving forecast accuracy.
- Created data visualizations and automated reporting dashboards using Matplotlib, Seaborn, and Power BI to communicate model performance metrics (AUC-ROC, precision/recall, F1-score) to stakeholders.
- Developed a real-time email monitoring application using Python to auto-parse incoming requests, classify query types, and generate reference tickets in ServiceNow, enabling 24/7 automated intake.
- Partnered with planning teams and business stakeholders to validate data consistency, support KPI tracking, and troubleshoot data anomalies across global operations.

Senior Systems Engineer | Infosys (India)

Jan 2017 - Dec 2019

- Automated data integration and ETL workflows using Python, Selenium, and SAP BW/APO, processing large-scale datasets across global supply chain systems and saving 800+ hours annually.
- Built structured data pipelines using SAP BW InfoCubes, DataSources, and process chains, ensuring clean, consistent data feeds for downstream statistical modeling and forecasting.
- Managed large volumes of master and transactional data across SAP APO and MM modules using CIF, performing data quality audits and validation checks to ensure accuracy across integrated systems.
- Wrote complex SQL queries for data extraction, transformation, and validation across SAP BW systems, supporting ad-hoc reporting for global supply chain operations.

- Supported demand and supply planning systems globally, resolving high-impact data issues through ServiceNow ticketing, monitoring batch jobs, and maintaining system uptime.
- Developed automated Python scripts for data reconciliation and anomaly detection to proactively identify data inconsistencies before downstream consumption.
- Partnered with planning teams to validate data consistency, troubleshoot data anomalies, and support forecasting accuracy across global operations.

PROJECTS

Slonik-7B-GRPO: Text-to-SQL Fine-Tuned LLM

Technologies: Python, PyTorch, QLoRA, GRPO, Qwen2.5-Coder-7B, BIRD-SQL, PostgreSQL, SQLite, GGUF, Ollama, HuggingFace

- Fine-tuned Qwen2.5-Coder-7B with QLoRA + GRPO (execution-based reinforcement learning) for PostgreSQL and SQLite text-to-SQL; achieved 38.20% on BIRD-PG outperforming GPT-4o (34.44%) at 7B parameters, 3x accuracy jump from 12.22% base model; published GGUF quantizations (Q4_K_M/Q5_K_M/Q8_0) for local deployment via Ollama and llama.cpp. [\[GitHub\]](#)

HuggingFace: huggingface.co/Phani-labs/Slonik-7B-GRPO

IntelliDesk: AI-Powered IT Service Desk

Technologies: Python, FastAPI, XGBoost, TF-IDF, Sentence-Transformers, FAISS, MLflow, Streamlit, Evidently AI, Docker, GitHub Actions

- Built an ML ticket-triage system with dual XGBoost classifiers for category and priority; diagnosed and fixed a 12:1 class imbalance where standard balanced weighting hurt F1, using sqrt-scaled weights, alongside error analysis and a sentence-transformer + FAISS retriever, all tracked in MLflow with Evidently drift monitoring and a 39-check validation suite. [\[GitHub\]](#)

DocBrain: Enterprise Document Intelligence Engine

Technologies: Python, PyTorch (CUDA), ChromaDB, Sentence-Transformers, Llama 3.1 8B, Tesseract/DocTR, SpaCy, FastAPI, Streamlit, Docker

- Built a grounded RAG pipeline over 4 document types with ChromaDB, Sentence-Transformers, and Llama 3.1 8B; ran a quantitative eval on 30 hand-written queries (0.90 Recall@5, 0.78 MRR, 1.00 Recall@1 on contracts) plus a second-layer LLM-judge faithfulness score, with quality-based OCR routing and SpaCy NER, 162 pytest tests. [\[GitHub\]](#)

EDUCATION

Master of Science - Electrical and Computer Engineering GPA: 3.61/4
Purdue University Northwest, Hammond, Indiana 2025
Coursework: Machine Learning, Neural Networks, Big Data, Computer Vision, Digital Signal Processing, Computer Networks

TECHNICAL SKILLS

Languages	Python, SQL, R, MATLAB, Bash, ABAP
ML / DL Frameworks	TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM, PySpark, Hugging Face Transformers
NLP & Computer Vision	NLTK, SpaCy, Sentence Transformers, OpenCV, FAISS, ChromaDB, Transfer Learning, OCR (Tesseract/DocTR)
LLM Fine-Tuning	QLoRA, GRPO, GGUF Quantization, Qwen2.5-Coder, Ollama, llama.cpp, HuggingFace Hub
Data Engineering	Pandas, NumPy, Apache Spark, Hadoop, ETL/ELT Pipelines, MySQL, Teradata, SAP BW/APO
Statistical Methods	Regression, Classification, Clustering, Time Series, A/B Testing, Hypothesis Testing, EDA, Dimensionality Reduction
Visualization & BI	Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau, Excel
MLOps & Deployment	FastAPI, Flask, Docker, Git, CI/CD, MLflow, Evidently AI, pytest, GitHub Actions, LangChain
Cloud & Platforms	AWS (S3, SageMaker, EC2), GCP (BigQuery, Vertex AI), Llama 3.1, ServiceNow

KEY ACHIEVEMENTS

- Fine-tuned Slonik-7B text-to-SQL model achieving 38.20% on BIRD-PG, outperforming GPT-4o (34.44%) at 7B parameters; 3x accuracy jump from 12.22% base model; published to HuggingFace with GGUF quantizations.
- Reduced IT ticket resolution time by 30% through automated NLP-based classification and knowledge retrieval, processing 200+ incidents per month.
- Saved 800+ hours annually by automating data integration, ETL workflows, and reporting pipelines using Python, Selenium, and SAP BW/APO.
- Built DocBrain RAG pipeline achieving 0.90 Recall@5, 0.78 MRR, 23ms p95 latency, 1.00 Recall@1 on contracts with 162 pytest tests and chunk-level citations.